

THE UNIVERSITY OF HONG KONG
DEPARTMENT OF MATHEMATICS
MATH3603 Probability theory

Tutorial 1 Solution

1. Show that $E \cap (F \cup G) = (E \cap F) \cup (E \cap G)$.

Solution:

“ $\supseteq$ ”: Since both $E \cap F$ and $E \cap G$ are subset of $E \cap (F \cup G)$, we have $E \cap (F \cup G) \supseteq (E \cap F) \cup (E \cap G)$.

“ $\subseteq$ ”: For $x \in E \cap (F \cup G)$, we have $x \in E$ and $x \in F \cup G$. The latter part means either $x \in F$ or $x \in G$. If $x \in F$ then $x \in E \cap F$; if $x \in G$ then $x \in E \cap G$. So either way, $x \in (E \cap F) \cup (E \cap G)$ and we are done.

2. Let E and F be mutually exclusive events in the sample space of an experiment with at least one of $P(E)$ and $P(F)$ being nonzero.
- (a) Suppose that the experiment is repeated until either event E or event F occurs. What does the sample space of this new super experiment look like?
- (b) Now we do not stop the experiment even if E or F occurs, and repeat it for infinitely many times. Show that the probability that event E occurs before event F is $\frac{P(E)}{P(E) + P(F)}$.

Hint: Argue that the probability that the original experiment is performed n times and E appears on the n -th time is $P(E) \times (1 - P(E) - P(F))^{n-1}$.

Solution:

If $P(E) = 0$ or $P(F) = 0$ then this question is trivial, so we can assume otherwise. Denote the event that neither E nor F happens as O . Then the sample space are $\{O^{n-1}E, O^{n-1}F\}$ where $n \in \mathbb{N}$. Since $P(O) = 1 - P(E) - P(F)$, we have $P(O^{n-1}E) = (1 - P(E) - P(F))^{n-1}P(E)$. Therefore the required probability is

$$\sum_{n=1}^{\infty} P(O^{n-1}E) = \sum_{n=1}^{\infty} (1 - P(E) - P(F))^{n-1} P(E) = \frac{P(E)}{1 - (1 - P(E) - P(F))} = \frac{P(E)}{P(E) + P(F)}$$

since $\sum_{n=1}^{\infty} (1 - P(E) - P(F))^{n-1}$ is an infinite geometric series with $0 < 1 - P(E) - P(F) < 1$.

- (a) Denote the event that neither E nor F happens as G . Since we always stop on the first occurrence of either E or F , the sample space of the new super experiment is

$$S = \{E, F, GE, GF, GGE, GGF, GGGE, GGGF, \dots\}.$$

- (b) Let $P(E) = p$ and $P(F) = q$ and suppose that the original experiment is performed without stopping. The probability that the original experiment is performed n times and E appears on the n -th time is

$$P(G^{n-1}E) = (1 - p - q)^{n-1}p.$$

Note that the events $\{G^{n-1}E\}_{n=1}^{\infty}$ are mutually disjoint, and their union is exactly the event that E occurs before F . Therefore, the probability that event E occurs before event F is

$$\begin{aligned} P(E \text{ occurs before } F) &= \sum_{n=1}^{\infty} P(G^{n-1}E) = \sum_{n=1}^{\infty} (1-p-q)^{n-1}p \\ &= p \sum_{n=0}^{\infty} (1-p-q)^n = \frac{p}{1-(1-p-q)} = \frac{p}{p+q} \\ &= \frac{P(E)}{P(E) + P(F)}. \end{aligned}$$

3. Let $\{E_i\}_{i=1}^{\infty}$ be a countable collection of events. Show that

$$P\left(\bigcup_{i=1}^{\infty} E_i\right) \leq \sum_{i=1}^{\infty} P(E_i),$$

where the right hand side is interpreted as infinite if the series diverges. This is known as Boole's inequality.

Hint: Define a collection of mutually disjoint events in a suitable way, then apply the countable additivity property of probability.

Solution:

The conclusion is trivial if the right hand side is infinite, so we assume that $\sum_{i=1}^{\infty} P(E_i) < \infty$. Define the events $F_1 = E_1$ and $F_i = E_i \setminus \left(\bigcup_{k=1}^{i-1} E_k\right) \subset E_i$ for $k \geq 2$. Note that $\{F_i\}_{i=1}^{\infty}$ is a collection of mutually disjoint events, since for any $i < j$ we have $x \in F_i \Rightarrow x \in E_i \Rightarrow x \in \bigcup_{k=1}^{j-1} E_k \Rightarrow x \notin F_j$. We claim that $\bigcup_{i=1}^{\infty} E_i = \bigcup_{i=1}^{\infty} F_i$ by showing inclusion from both sides.

- “ $\supseteq$ ”: If $x \in \bigcup_{i=1}^{\infty} F_i$, then $x \in F_j$ for some $j \in \mathbb{N}$. By definition of F_j , we have $x \in E_j$, so $x \in \bigcup_{i=1}^{\infty} E_i$.
- “ $\subseteq$ ”: Assume that $x \in \bigcup_{i=1}^{\infty} E_i$. Take the smallest $j \in \mathbb{N}$ such that $x \in E_j$. If $x \notin F_j = E_j \setminus \left(\bigcup_{k=1}^{j-1} E_k\right)$, then we must have $x \in \bigcup_{k=1}^{j-1} E_k$ and hence $x \in E_k$ for some $k \leq j-1$, contradicting the minimality of j . Thus $x \in F_j$, and hence $x \in \bigcup_{i=1}^{\infty} F_i$.

By countable additivity of probability, we have

$$P\left(\bigcup_{i=1}^{\infty} E_i\right) = P\left(\bigcup_{i=1}^{\infty} F_i\right) = \sum_{i=1}^{\infty} P(F_i).$$

By the above equality, we have $\sum_{i=1}^{\infty} P(F_i) \leq 1$, so both $\sum_{i=1}^{\infty} P(E_i)$ and $\sum_{i=1}^{\infty} P(F_i)$ converge. Since $F_i \subseteq E_i$, we have $P(F_i) \leq P(E_i)$ for each $i \in \mathbb{N}$, and hence $\sum_{i=1}^{\infty} P(F_i) \leq \sum_{i=1}^{\infty} P(E_i)$. Therefore,

$$P\left(\bigcup_{i=1}^{\infty} E_i\right) \leq \sum_{i=1}^{\infty} P(E_i).$$

4. There are n people in a room and each of them owns a hat. Now each person throws his hat into the center of the room, and the hats are then randomly returned to the people so that each person receives one hat. Assume that all possible assignments of hats to people are equally likely.

- (a) If $n = 2$, what is the probability that no person gets his own hat back?
- (b) For general n , what is the probability that no person gets his own hat back?

Solution:

- (a) When $n = 2$, there are only two possible assignments of hats: either each person gets his own hat back, or no person gets his own hat back. Since these two assignments are equally likely, the probability that no person gets his own hat back is $\frac{1}{2}$.
- (b) For general n , let X be the whole sample space and A_i be the event that the i -th person gets his own hat back, for $i = 1, 2, \dots, n$. The event that no person gets his own hat back is $X \setminus (\bigcup_{i=1}^n A_i)$. By the principle of inclusion-exclusion, we have

$$\begin{aligned} P\left(\bigcup_{i=1}^n A_i\right) &= \sum_{k=1}^n (-1)^{k+1} \left(\sum_{1 \leq i_1 < i_2 < \dots < i_k \leq n} P(A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_k}) \right) \\ &= \sum_{i_1=1}^n P(A_{i_1}) - \sum_{1 \leq i_1 < i_2 \leq n} P(A_{i_1} \cap A_{i_2}) + \dots + (-1)^{n+1} P(A_1 \cap A_2 \cap \dots \cap A_n). \end{aligned}$$

For each $1 \leq k \leq n$, consider the event that k specific people $i_1, \dots, i_k$ get their own hats back, i.e., the event $A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_k}$. The number of ways to assign the remaining $n - k$ hats to the remaining $n - k$ people such that each person receives one hat is $(n - k)!$. Since the total number of ways to assign n hats to n people is $n!$, we have

$$P(A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_k}) = \frac{(n - k)!}{n!}, \quad \forall 1 \leq i_1 < i_2 < \dots < i_k \leq n.$$

Hence, we have

$$\begin{aligned} P\left(\bigcup_{i=1}^n A_i\right) &= \sum_{k=1}^n (-1)^{k+1} \left(\sum_{1 \leq i_1 < i_2 < \dots < i_k \leq n} \frac{(n - k)!}{n!} \right) \\ &= \sum_{k=1}^n (-1)^{k+1} \left(\frac{(n - k)!}{n!} \cdot T_{n,k} \right), \end{aligned}$$

where $T_{n,k}$ denotes the number of terms in the summation $\sum_{1 \leq i_1 < i_2 < \dots < i_k \leq n}$, or equivalently, the number of ordered tuples $(i_1, \dots, i_k)$ with $i_1 < i_2 < \dots < i_k$ in $\{1, \dots, n\}$. Note that $T_{n,k}$ is simply the number of ways to choose k people from n people (why?), which is given by the binomial coefficient $\binom{n}{k} = \frac{n!}{k!(n - k)!}$. Hence, we have

$$\begin{aligned} P\left(\bigcup_{i=1}^n A_i\right) &= \sum_{k=1}^n (-1)^{k+1} \left(\frac{(n - k)!}{n!} \cdot \frac{n!}{k!(n - k)!} \right) \\ &= \sum_{k=1}^n \frac{(-1)^{k+1}}{k!}. \end{aligned}$$

Therefore, the probability that no person gets his own hat back is

$$P\left(X \setminus \left(\bigcup_{i=1}^n A_i\right)\right) = 1 - P\left(\bigcup_{i=1}^n A_i\right) = 1 - \sum_{k=1}^n \frac{(-1)^{k+1}}{k!} = \sum_{k=0}^n \frac{(-1)^k}{k!}.$$

Remark: A permutation of $\{1, 2, \dots, n\}$ is called a derangement if no element appears in its original position. The above problem is equivalent to finding the probability that a random permutation of $\{1, 2, \dots, n\}$ is a derangement. In particular, the above probability implies that the number of derangements of $\{1, 2, \dots, n\}$ is given by $n! \sum_{k=0}^n \frac{(-1)^k}{k!}$.